

EMO: A Plug-in Emission Monitor For Vehicles

Using the EMO device from Logica, drivers can reduce their vehicle emissions by up to 15 per cent simply by improving the way they drive. Apart from reduced emissions, they also benefit from fuel savings since better driving habits lead to improved vehicle mileage



The EMO measures carbon emissions from a vehicle and transmits the information to a central server (Image courtesy: <http://business.outlookindia.com>)

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Pollution, in particular, vehicular pollution, is a major concern today. There are over 600 million passenger vehicles in the world accounting for thousands of tonnes of carbon emission every single day. However, most of the solutions to combat vehicle pollution, such as electric cars and bio-fuels, require either drastic changes in social infrastructure or major investment from vehicle owners. As a result, their adoption is slow.

Another way to counter pollution would be to target its source. One major reason for vehicular pollution is not just the vehicles we drive but how we drive them. Poor driving habits like aggressive driving, engine idling and over-speeding have a large impact on fuel efficiency and thus harmful gas emissions from the vehicle. This is

where Logica's EMO comes in. The device helps reduce vehicle emissions by monitoring how people drive and then providing inputs to them to improve their driving styles.

What is it?

The device is installed in the vehicle and connected to the vehicle's onboard computer. It connects to the OBD-II port, which is available in most vehicles today, and allows access to the data from the engine control unit (ECU).

Once connected, the device monitors in real time various driving behaviour related parameters such as speed, accelerations and braking. Using the data gained, the on-board computer determines the vehicle's carbon emissions. The device has a SIM card inside which is used to send all the computed data to Logica backend servers using

GPRS, where it is analysed and a report is generated.

The report thus made is shared with the driver through numerous ways such as a Web interface, SMS alerts, smartphone apps and e-mail reports. The analysis from the report helps him understand how he is currently driving versus how he should be driving and thus improve his driving habits.

Sanjoy Ghosh, product manager at Logica, claims that by using the EMO, drivers can reduce their vehicle emissions by up to 15 per cent simply by improving the way they drive. Apart from reduced emissions, drivers also benefit from fuel savings since better driving habits lead to improved vehicle mileage.

Hardware design challenges

It took the team about three years to develop the Logica EMO. "We needed to create a safe, efficient and reliable system to measure emissions in real time. This required understanding of the technology used in latest vehicles and how to extract data from them in a safe and reliable way. In addition, the device had to undergo extensive testing and certification before it could be offered to clients. The solution today is CE certified and has also been independently audited by a leading carbon footprint company, Zerofootprint, for accuracy," says Sanjoy.

The hardware of the device is compliant with all the latest models of vehicles having a standard OBD-II protocol port. Moreover, it has telephony for GPRS-based wireless data transmis-